



Ultimate Stat Lab

User Guide

Explore, compare, and discover Pokémon data
through interactive charts and filters.



<https://datadex.famruiz.com/>

1 Getting Started

Data-Dex is an interactive browser-based dashboard designed to introduce simple data science concepts to kids using a subject they already love — Pokémon. A filter sidebar runs down the left side; the main area holds six analysis cards. Every filter change instantly refreshes all cards — no submit button required.

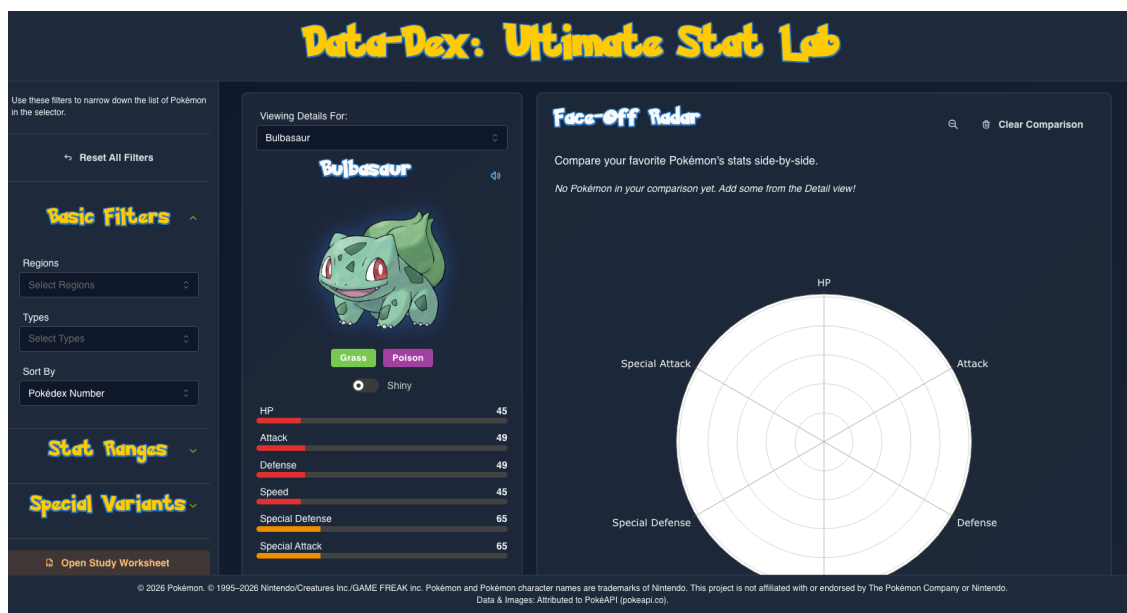
The app comes with two companion PDFs to support classroom use:

- Student Worksheet — a guided activity sheet with questions for students to fill in while exploring the app. A download button is also available at the bottom of the filters sidebar.

datadex.famruiz.com/worksheet

- Instructor Cheat Sheet — suggested answers and teaching notes for educators.

datadex.famruiz.com/cheat-sheet



Full app overview — filters sidebar (left) + analysis cards (main area)

2 Filters Sidebar

The sidebar controls which Pokémon appear across every card simultaneously. It has three collapsible sections:

Basic Filters

- **Regions**
Restrict to one or more regional Pokédexes (Kanto, Johto, Hoenn...).
- **Types**
Show only Pokémon whose primary or secondary type matches.
- **Sort By**
Order the selector list by Pokédex # or alphabetically.

Stat Ranges

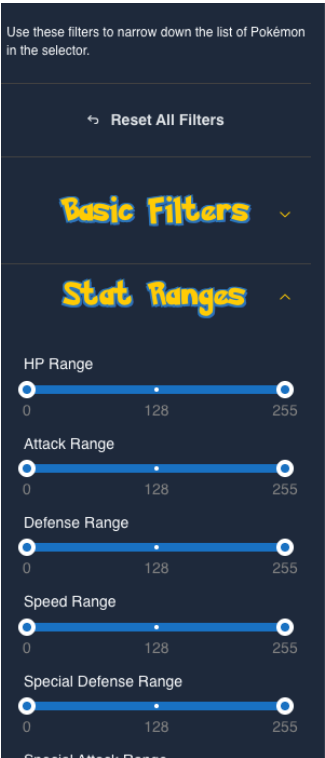
- **HP / Atk / Def / Sp.Atk / Sp.Def / Speed**
Each stat has a 0–255 range slider. Only Pokémon within all ranges appear.

Special Variants

- **Mega / Regional / G-Max**
Toggle ON/OFF to include or exclude these forms.
- **Legendaries / Mythicals / Ultra Beasts**

Toggle ON/OFF for each category.

- Final Evolutions Only**
When ON, hides unevolved and mid-stage Pokémon.
- Reset All Filters**
Restores every filter and toggle to its default state.



Expanded filters sidebar showing all three sections

One filter change instantly updates the Detail Card selector, Type Leaderboard, and World Exploration scatter plot. The Face-Off Radar is unaffected — Pokémon already added to it stay on the chart regardless of filters.

3 Pokémon Detail Card

The Detail Card sits in the upper-left of the main area. Use it to look up any Pokémon and add them to the Face-Off Radar.

■ Pokémon Selector

Search and pick any Pokémon from the filtered list.

■ Sprite / Shiny

Official artwork. Toggle Shiny for alternate colours.

■ Type Badges

Shows primary and secondary type(s).

■ Stat Bars

HP, Atk, Def, Sp.Atk, Sp.Def, Speed with colour coding.

■ Play Cry

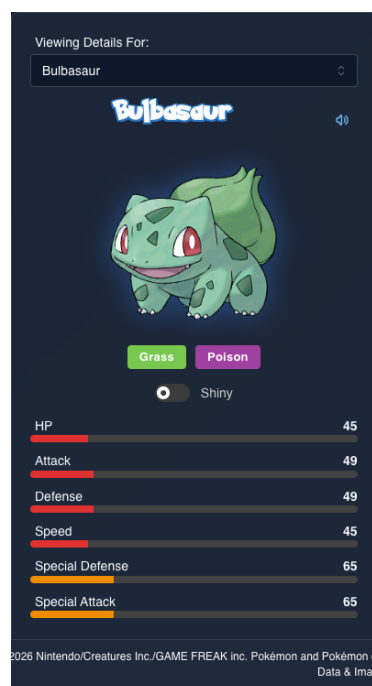
Speaker icon plays the Pokémon's in-game sound.

■ Add to Comparison

Adds the current Pokémon to the Face-Off Radar (max 6).

■ Evolution Lineage

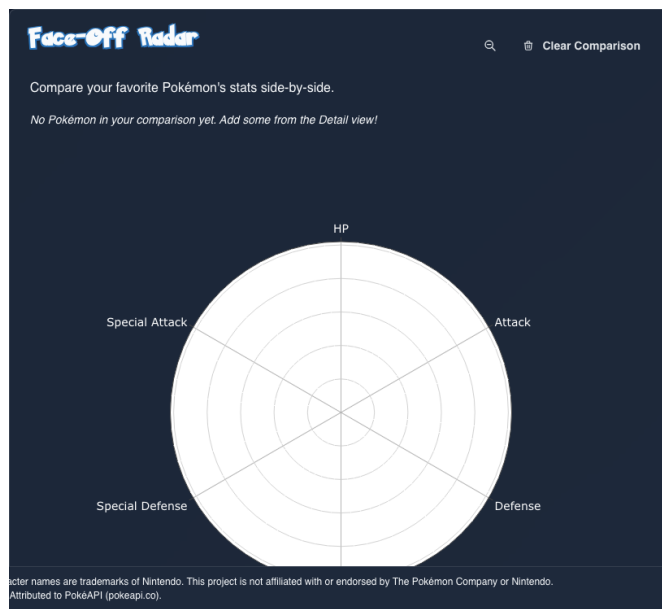
Tap a sprite in the chain to jump to that Pokémon.



Pokémon Detail Card — Bulbasaur selected

4 Face-Off Radar

A spider/radar chart that overlays base stats for up to 6 Pokémon at once. Select a Pokémon in the Detail Card, click Add to Comparison, and repeat. Hover any spoke tip to see the exact stat value. Scroll to zoom in; use the Reset Zoom button to return to default. Click Clear Comparison to remove all Pokémon from the chart.



Face-Off Radar — add Pokémon via the Detail Card to populate this chart

Note: sidebar filters affect which Pokémon you can add via the selector, but any Pokémon already on the radar will stay there even if a filter would exclude them. Use Clear Comparison to remove them manually.

5 Trainer Comparison

Enter your height (ft) and weight (lbs) to see how you physically compare with the currently selected Pokémon. Results update automatically as you type.

Your Height (ft)

Valid range 2–7 ft. Auto-updates after a short pause.

Your Weight (lbs)

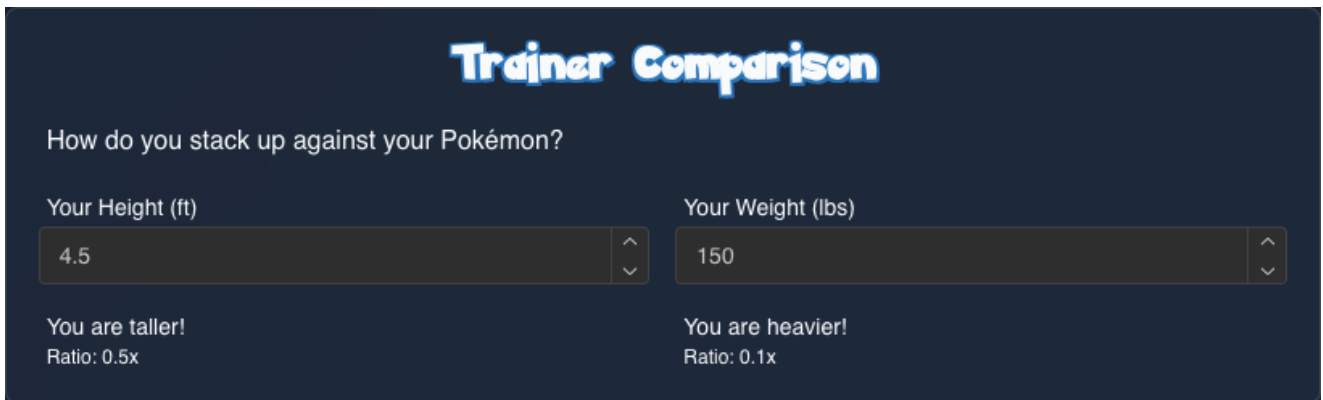
Valid range 10–500 lbs. Auto-updates after a short pause.

Height result

States who is taller and the ratio (e.g. 'Charizard is taller — 1.2×').

Weight result

States who is heavier and the ratio (e.g. 'You are heavier — 1.6×').

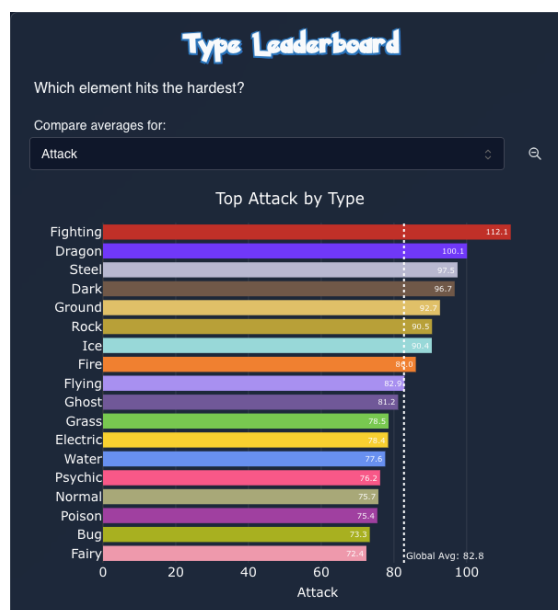
The image shows a dark-themed web interface titled "Trainer Comparison" in a stylized blue and white font. Below the title, it asks "How do you stack up against your Pokémon?". There are two input fields: "Your Height (ft)" with the value "4.5" and "Your Weight (lbs)" with the value "150". Below each field, there is a result: "You are taller! Ratio: 0.5x" and "You are heavier! Ratio: 0.1x".

Input	Value	Result
Your Height (ft)	4.5	You are taller! Ratio: 0.5x
Your Weight (lbs)	150	You are heavier! Ratio: 0.1x

Trainer Comparison — height and weight inputs with live comparison results

6 Type Leaderboard

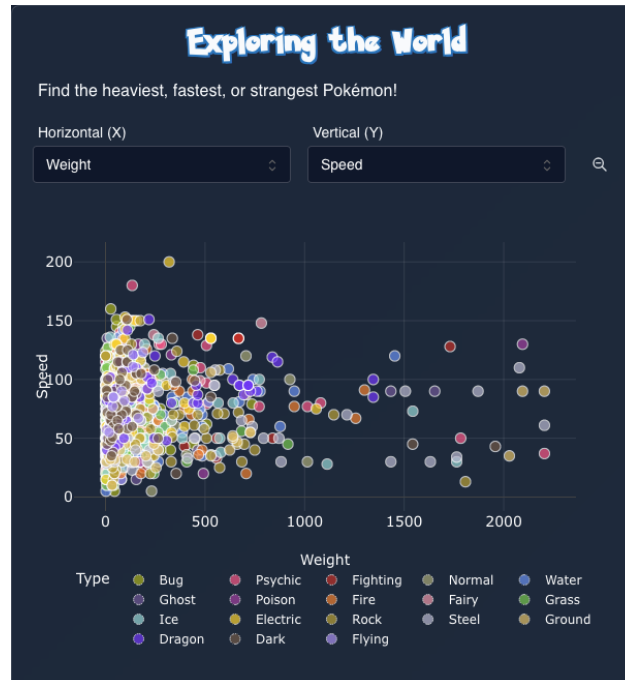
Horizontal bar chart ranking all Pokémon types by average base stat. Use the dropdown to switch between HP, Attack, Defense, Sp. Atk, Sp. Def, or Speed. Hover a bar for the exact average. Sidebar filters restrict the Pokémon pool — try selecting a single region to see how types rank within one generation.



Type Leaderboard — ranked by average Attack stat across all Pokémon

7 Exploring the World

An open-ended scatter plot where you choose any two axes. Both the horizontal (X) and vertical (Y) axes can be set independently to any of the six base stats, Height, or Weight. Each dot represents a single Pokémon, coloured by its primary type. Hover a dot to see its name and exact values. Scroll to zoom, drag to pan, and use the Reset Zoom button to restore the default view. Sidebar filters instantly narrow the dataset.



Exploring the World — Weight (X) vs Speed (Y) reveals outliers like Celesteela and Shuckle

Try Height vs Weight to spot the heaviest Pokémon, or Speed vs HP to find fast, bulky Pokémon like Regieleki.

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